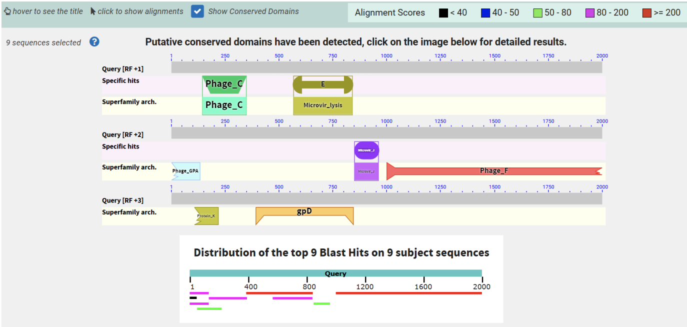
Task 1. Fungal opsins

Have you found any significant matches? How do findings correspond to 1e-5 e-value cutoff (findings with higher e-value may be surely viewed as non-homologous) and 1e-25 e-value cutoff (finding with lower e-value can be viewed as homologous)?	There are a lot of significant matches lower than 1e-5 e-value cutoff. And two matches below 1e-25 cutoff.
Name one sequence that is in top-10 similar sequences by score.	XP_040722269.1
How many identical residues are found in the alignment of the query and this hit?	87
Name the organism that produced this hit.	Protomyces lactucae-debilis
Look up this fungus, how does it live? Is it more similar to champignons (multicellular fungi) or yeast (mostly unicellular, microscopic fungi) or probably, mold? Give a link to the information source.	Dimorphic / yeast-like plant parasite. It's a **dimorphic ascomycete**: yeast-like outside, hyphae + spores inside host tissue. Infection of suitable host tissue → intercellular mycelium. Induced hypertrophy / gall-like structures where sexual spores develop. Resting/ascospores germinate back into yeast-like cells that spread and wait for new hosts. https://studiesinmycology.org/sim/Sim30/fulltext/index.html?i=12

Task 2. Phage fragments

Fragment identifier from phage_fragments.fasta that you used	phage_1_fragment_0
Blast type you used to identify origin genome (1)	Blastn megablast
Phage species name	Escherichia phage phiX174
Which metrics tell you that you have found the source genome?	E value = 0.0 Percent Identity = 100% Query coverage = 100% Bit score = 3694
Blast type you used to identify encoded virus proteins (2) Specify blast type and options of the search that you used.	Blastx, refseq_protein database, organism Escherichia phage phiX174 (taxid:10847) compared to: Blastx, nr database with same organism restriction
What are probable encoded proteins? Read the names of findings and name the probable proteins similarly — protein may have a name “protein A”. How many proteins and/or protein parts are encoded in this piece of viral DNA? Focus on the viral proteins.	protein C, protein E, protein F, protein J, protein A*, protein D, protein K, protein B, protein A 9 proteins and/or protein

Insert the picture from the graphic summary tab.	

Task 3. Dino in journal

I will not allow the publication to appear, because Blastn showed that this sequence has a lot of significant alignments with different cloning vectors. E value = 1e-116. Query coverage > 90%, indicating that it is a modern engineered DNA.